MD HAFIZUR RAHMAN

Sakellaris Ph.D. Fellow | Electrical & Computer Engineering
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Google Scholar: 518 Citations | **h-index:** 10 | **i10-index:** 10

Education

- **Ph.D. in Electrical & Computer Engineering** — January 2023 - Present
  University of Maine, Orono, USA — *Research Assistant - SIEGE Lab*
- **Graduate Certificate in Data Science & Engineering** — May 2026
  University of Maine, Orono, USA
- **M.S. in Computer Engineering** — January 2023 - August 2025
  University of Maine, Orono, USA
- **B.Sc. in Electrical & Electronic Engineering** — January 2016 - May 2022
  Islamic University, Bangladesh

Patent Applications

1. **METHODS AND SYSTEMS FOR THE DESIGN AND USE OF INTELLIGENT LAYER SHARED NEURAL NETWORKS AND MODEL ARCHITECTURE SEARCH ALGORITHMS**, International Patent Appln. No.: PCT/US24/44769, Prabuddha Chakraborty, Sumaiya Shomaji, Md Hafizur Rahman, Md Mashfiq Rizvee
2. **METHODS AND SYSTEMS FOR THE AUTOMATED GENERATION OF NEURAL NETWORK ARCHITECTURES**, International Patent Appln. No.: PCT/US2025/013847, Prabuddha Chakraborty, Md Hafizur Rahman, Zafaryab Haider
3. **FRAMEWORK FOR MITIGATING MALICIOUS RLHF FEEDBACK IN LLM TRAINING USING CONSENSUS BASED REWARD**, International Patent Appln. No.: PCT/US26/18361, Zafaryab Haider, Md Hafizur Rahman, Shane Moeykens, Prabuddha Chakraborty
4. **BIT-FLIP INDUCED SEMANTIC DRIFT IN MULTIMODAL TRANSFORMER MODELS**, U.S. Provisional Patent Application No. 63/930,131, Zafaryab Haider, Md Hafizur Rahman, Shane Moeykens, Prabuddha Chakraborty
5. **MEASURING HARM AMPLIFICATION IN MULTI-AGENT LLM SYSTEMS**, U.S. Provisional Patent Appln. No.: 64/148,848, Prabuddha Chakraborty, Zafaryab Haider, Tanzim Mahfuz, Md Hafizur Rahman

Selected Peer-Reviewed Research Articles

1. **SMASH: Probing Speech Recognition Robustness via Semantically Targeted Bit Flips**, Zafaryab Haider, Md Hafizur Rahman, Aysegul Bumin, & Prabuddha Chakraborty, ***Accepted at NeurIPS, 2026.***
2. **Intelligent Layer Sharing (ILASH): A Predictive Neural Architecture Search Framework for Multi-Task Applications**, Md Hafizur Rahman, Zafaryab Haider, Md Mashfiq Rizvee, Sumaiya Shomaji, & Prabuddha Chakraborty, ***IEEE Access, 2025.***
   Impact Factor: 4.2
3. **An automated multi parameter neural architecture discovery framework using ChatGPT in the backend**, Md Hafizur Rahman, Zafaryab Haider, & Prabuddha Chakraborty, ***Scientific Reports, 2025.***
   Impact Factor: 4.9
4. **A framework for mitigating malicious RLHF feedback in LLM training using consensus based reward**, Zafaryab Haider, Md Hafizur Rahman, Vijay Devabhaktuni, Shane Moeykens & Prabuddha Chakraborty, ***Scientific Reports, 2025.***
   Impact Factor: 4.9
5. **TRIM: AI Guided Random Number Generation for Resource-Constrained IoT Systems**, Tasneem Suha, Md Hafizur Rahman, Ashley E Rice, Colin Smith, Rima Asmar Awad, & Prabuddha Chakraborty, ***IEEE Access, 2025.***
   Impact Factor: 4.2
6. **In-situ particle analysis with heterogeneous background: a machine learning approach**, Adeeb Ibne Alam, Md Hafizur Rahman, Akhter Zia, Nate Lowry, Prabuddha Chakraborty, Md Rafiul Hassan & Bashir Khoda, ***Scientific Reports, 2024.***
   Impact Factor: 4.9

7. **Real-time face mask position recognition system based on MobileNet model**, Md Hafizur Rahman, Mir Kanon Ara Jannat, Md Shafiqul Islam, Giuliano Grossi, Sathya Bursic & Md Aktaruzzaman, ***Smart Health, Elsevier, 2023.***
   Impact Factor: 3.4

8. **RATNet: A deep learning model for Bengali handwritten characters recognition**, Md Shafiqul Islam, Md Moklesur Rahman, Md Hafizur Rahman, Massimo Walter Rivolta, Md Aktaruzzaman, ***Multimedia Tools and Applications, Springer, 2022.***
   Impact Factor: 3.6

9. **CSI-IANet: An Inception Attention Network for Human-Human Interaction Recognition Based on CSI Signal**, M Humayun Kabir, M Hafizur Rahman, Wonjae Shin, ***IEEE Access, 2021.***
   Impact Factor: 4.2

Poster Presentations & Talks

- **LEMONADE: Large Language Models for Task-Specific Neural Network Design** *April 17, 2026*
  University of Maine Student Symposium-2026
- **ILASH: Predictive Neural Architecture Search for Efficient Multi-Task Layer Sharing** *April 17, 2026*
  University of Maine Student Symposium-2026
- **An automated multi-parameter neural architecture discovery framework using ChatGPT in the backend** University of Florida, USA
  Warren B. Nelms Annual IoT Conference *4–5 December, 2025*
- **ILASH: A Predictive Neural Architecture Search Framework for Multi-Task Applications** University of New Hampshire, USA
  WHiSPerS: Wearable Health monitoring Sensors for Privacy and Security *21 March, 2025*

Honors, Awards & Fellowships

- **George P. Sakellaris'69 & Caterina Papoulias-Sakellaris Graduate Fellowship**
  University of Maine, College of Engineering, 2026–2027 academic year *Sept 1, 2026 - Aug 31, 2027*
- **MIRTA 8.0 Commercialization Accelerator Award - *(Co-Investigator)*** *January 14–May 11th, 2026*
  Awarded MIRTA funding for customer discovery to commercialize the COBRA invention (Int. Patent Application No. PCT/US2025/013847) *Amount: $25,000*
- **Travel Grant, Bangladesh Sweden Trust Fund** *2023*
  Given by Ministry of Finance, Bangladesh.
- **University Merit Scholarships** *2017 & 2018*
  Given by Islamic University, Bangladesh.

Academic Service

- **Peer Reviewer (50 Manuscript Reviews):** IEEE Access (41), IEEE Internet of Things Journal (4), Multimedia Tools and Applications (2), Journal of Big Data (1), International Journal of Machine Learning and Cybernetics (1), Applied Artificial Intelligence (1).
- **Undergraduate Research Judge (2024–2026):** University of Maine Student Symposium.

Research Experience

- **Secure and Intelligent Edge Computing (SIEGE) Lab** University of Maine, Orono, USA
  Research Assistant *Jan 2023–Present*
  - **Director**: Dr. Prabuddha Chakraborty, Dr. Waldo "Mac" Libbey '44 Professor & Assistant Professor, Electrical & Computer Engineering, University of Maine.
  - **Neural Architecture Search**: Developed ILASH, a neural architecture search framework for efficient multi-task learning, resulting in a peer-reviewed publication and contributing to international patent applications.
  - **AI Security**: Conducted research on LLM security, including malicious RLHF feedback, multi-agent harm amplification, and bit-flip attacks, contributing to peer-reviewed publications, including NeurIPS 2026.

Research Interests & Technical Skills

- **Deep Learning & ML Toolkits**: TensorFlow, Keras, PyTorch, Pandas, Matplotlib, Seaborn, GNN, LangChain, LangGraph, MCP, RAG
- **Languages & Systems**: Python, MATLAB, C, C++, Windows, Debian
- **Hardware/Edge Devices**: Raspberry Pi, Arduino, NVIDIA Jetson Nano, Chipwhisperer, ESP-32, ESP8266 Node MCU

Professional Certifications

- Building AI Agents with Multimodal Models (NVIDIA)
- Deep Learning Specialization
- AI for Everyone
- Introduction to the Internet of Things and Embedded Systems

Membership in Scientific Societies

IEEE (Graduate Student Member, Jan 2021–Present) | **ACM** (Student Member, Feb 2025–Present)